

ERP Solution for Digital Enterprises: SD, MM & PP Implementation for a Private Hospital Management System

by

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A PBL report for the ERP Solution for Digital Enterprises



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Introduction:

Nowadays, hospitals need all their departments to work together properly, starting from patient registration to treatment, pharmacy, and reporting. But in many private hospitals, departments still work separately. Because of this, data gets scattered, patient care can be delayed, and overall work becomes inefficient.

To solve these issues, we developed an SAP ERP-based solution specially designed for hospital management. In our system, we used three main modules — **Sales and Distribution (SD)**, **Materials Management (MM)**, and **Production Planning (PP)** — and combined them into one platform.

Here, patients are treated as customers using the SD module. The MM module is used to manage the purchase of medicines and medical equipment. The PP module is used for planning and handling lab tests and diagnostic activities.

By connecting all these modules, the system brings everything into one place. This reduces manual work, avoids entering the same data multiple times, and helps hospital staff make quicker and better decisions.

In this report, we have explained the overall system, including the organizational structure, master data, and how each module works. We also showed how these modules are connected and how they help in improving hospital operations.

About the Hospital:

The subject of this ERP implementation is a mid-to-large-scale private hospital operating in a multi-specialty environment. The hospital provides a comprehensive range of healthcare services including outpatient and inpatient care, diagnostic laboratory services, pharmacy operations, surgical procedures, and specialized treatment programs.

Prior to ERP implementation, the hospital's departments — including patient admissions, pharmacy, billing, finance, and human resources — operated independently with minimal data sharing. This fragmented approach resulted in medication stockouts, billing discrepancies, delayed approvals, and inconsistent patient records. The lack of an integrated system created operational bottlenecks that directly impacted patient care quality and institutional efficiency.

The hospital recognized that a technology-driven transformation was essential to remain competitive and deliver high-quality, patient-centric care. This led to the decision to implement SAP ERP as the backbone of its digital transformation strategy.

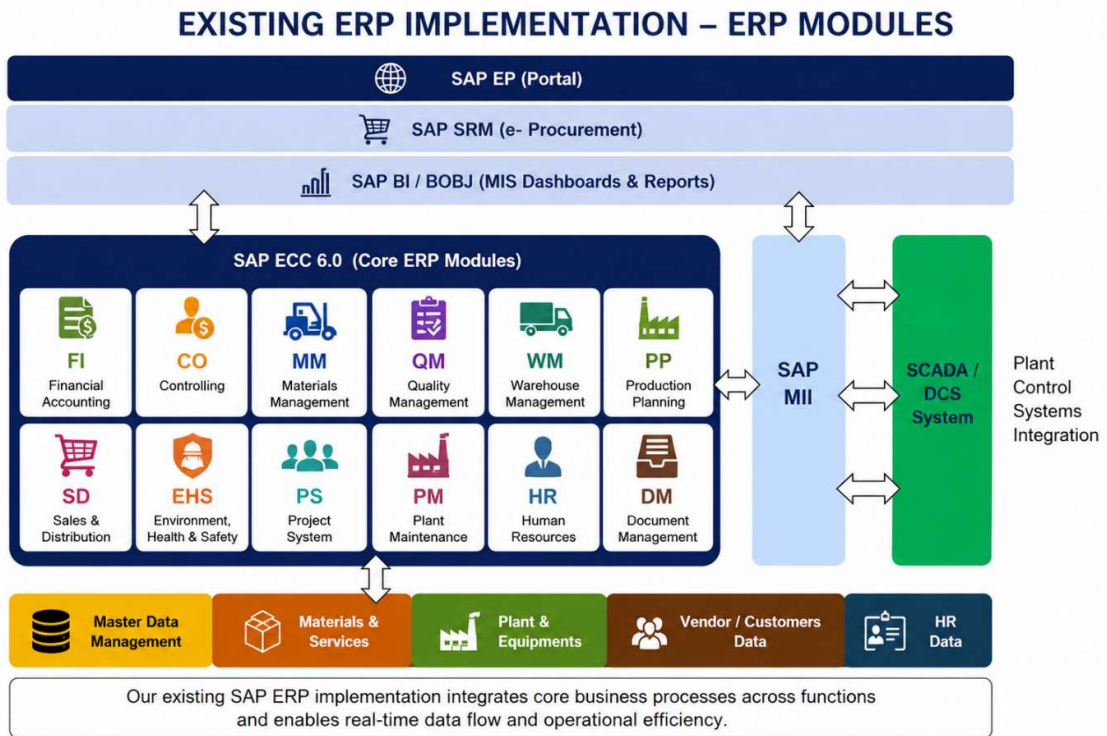
Existing ERP Implementation:

The hospital chose SAP ERP to automate and standardize processes across all its departments and physical locations. The implementation was designed to establish an integrated business application architecture — an ERP backbone that supports day-to-day hospital functioning and drives the institution's long-term strategic goals.

Key implementation details:

- SAP Version: ECC 6.0
- SAP Modules Implemented: SD, MM, PP, FICO, HR
- Scope: End-to-end hospital operations including patient services, pharmacy, lab, billing, finance, and staff management

- Integration: Full cross-module data flow — patient registration through billing and financial posting
- Data Architecture: Centralized master data for patients, medicines, vendors, and staff with a single source of truth



"Our SAP-based hospital ERP has laid down the foundation for sustainable healthcare reforms by integrating and enabling core hospital operations with information systems at the transaction level."

Key Challenges before SAP ERP was implemented

Before implementing SAP ERP, the hospital faced multiple challenges across its operations due to limited adoption of information technology and fragmented processes. The key challenges were:

- Independent departments with no shared data, no coordination, and no unified platform
- Delayed patient care due to manual registration, paper-based records, and slow information flow
- Frequent medicine stockouts caused by the lack of real-time inventory visibility across the pharmacy
- Billing errors and revenue leakage arising from disconnected billing and financial systems
- Financial mismanagement due to manual tracking of payments, expenses, and budget allocation
- Lack of visibility on medicine and equipment inventory across departments
- Each department maintained its own data for staff and patient records manually
- Difficulty in managing procurement processes — vendor selection, purchase orders, and goods receipt were paper-based

- Absence of real-time lab scheduling and diagnostic workflow management
- Inability to generate consolidated financial and operational reports across departments

Reasons for Choosing SAP

After a thorough evaluation of available ERP platforms, SAP was selected due to the following strategic reasons:

- SAP is a comprehensive system with most of the functionality required for healthcare operations with seamless cross-module integration
- It provides a robust framework to enhance functionality to suit specific business requirements of the hospital
- SAP is a product well-suited to service-based environments with required localizations and compliance features
- Several large hospital chains and healthcare institutions are already successfully using SAP
- SAP ASAP methodology ensures structured implementation within defined timelines
- Business Process mapping: All stakeholder departments were involved to define the To-Be processes
- Change Management: User awareness and acceptance ensured through workshops, demos, and training sessions
- Master Data Management: Teams were formed to collect, clean, format, and validate patient, vendor, and material data
- Training: Project team, power users, and end users were all trained before go-live

Proposed Implementation:

The hospital successfully modernized its core operations by implementing SAP ECC 6.0 across all departments. While the ERP has stabilized operations and provided a centralized data platform, the current interface is restricted to desktop environments and traditional SAP GUI transactions.

To further enhance operational efficiency — especially for bedside nursing, field-based diagnostics, and remote patient monitoring — the hospital proposes the implementation of SAP Fiori. This shift will transition the hospital ERP from a transaction-heavy desktop system to an intuitive, role-based, and mobile-ready digital workplace.

Business Drivers for Fiori Adoption

Building on the successes of the ECC implementation, the following challenges necessitate the move to Fiori:

- Bedside Mobility: Nursing and clinical staff currently rely on paper-based notes before entering data into SAP at a desk
- Approval Bottlenecks: Department heads require real-time approval capabilities for procurement and budget requests while on the move
- User Experience: All hospital users navigate complex SAP GUI screens; a simplified UI will reduce training costs and data entry errors
- Decision Support: Transitioning BI insights into executive dashboards accessible via tablets and smartphones for real-time operational visibility

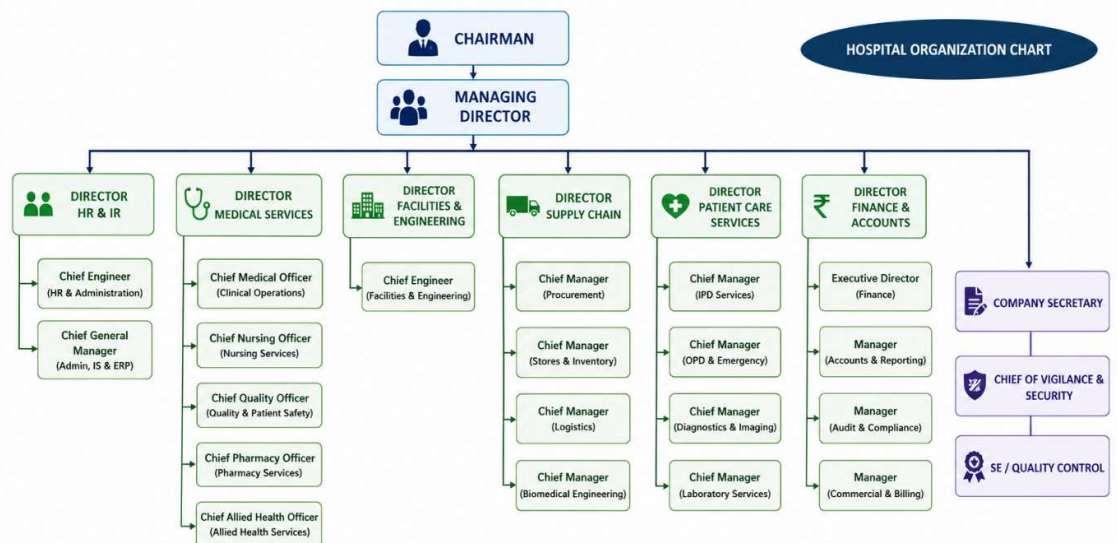
Expected Strategic Benefits

"The transition to SAP Fiori represents the next evolution of our hospital ERP — moving from a system of record to a system of engagement."

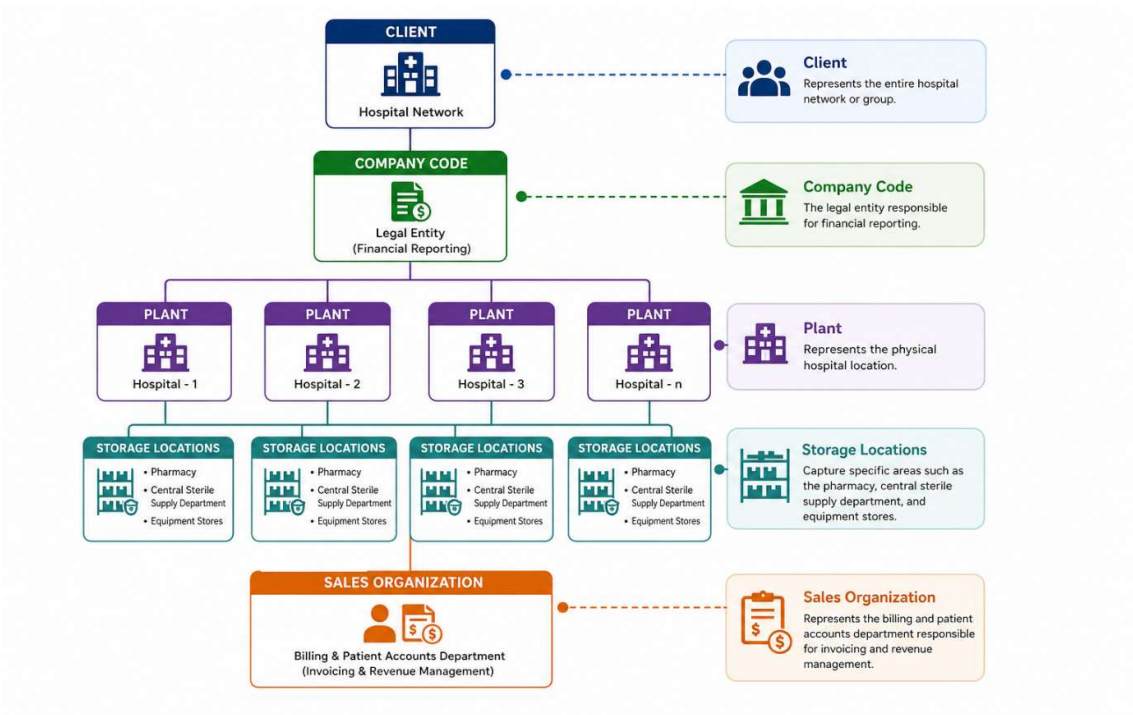
- Improved Patient Safety: Faster medication dispensing and treatment recording ensures clinical accuracy and patient safety
- Higher Staff Productivity: Reduction in administrative desk-time for clinical staff, freeing them for patient care
- Better Data Integrity: Real-time data entry at the point of care reduces reconciliation efforts and errors
- Cost Savings: Faster procurement approval cycles lead to better vendor relationships and optimized inventory costs

Organizational Structure

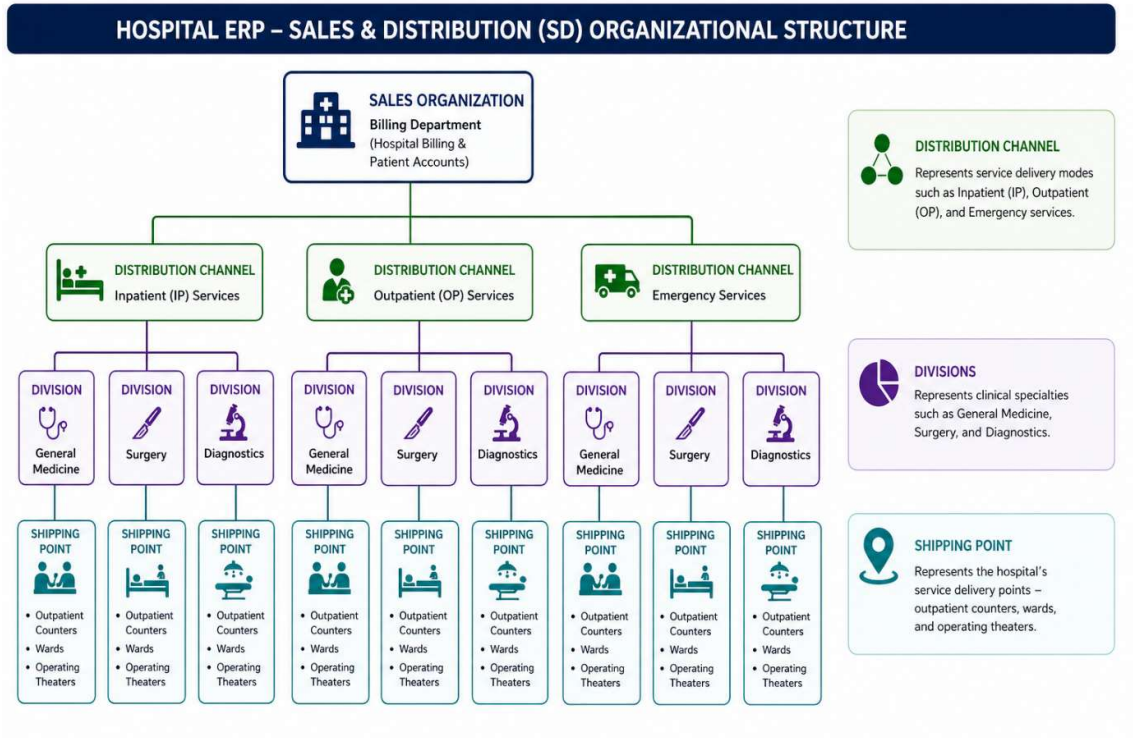
HOSPITAL – OVERALL ORGANIZATION STRUCTURE



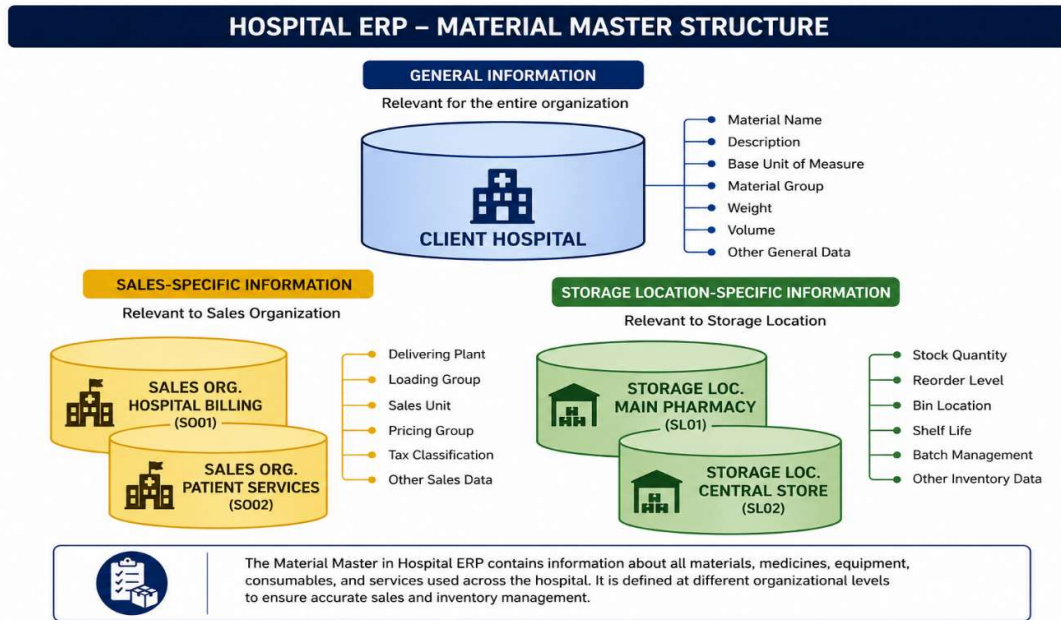
Material Management Organization Structure



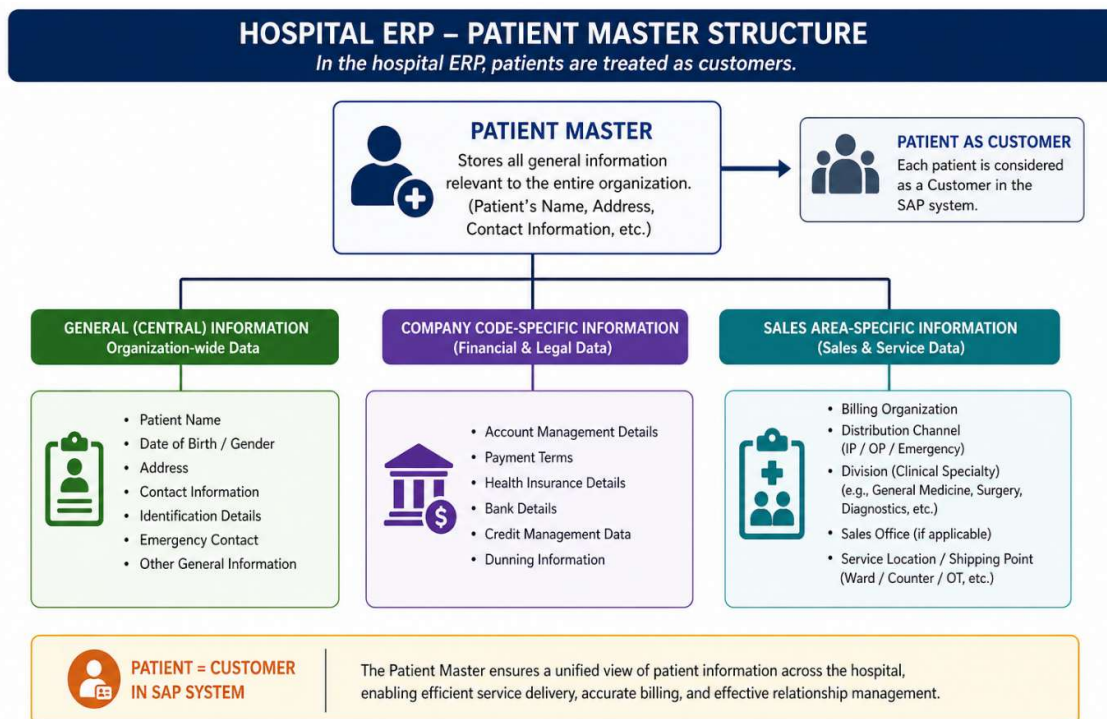
Sales and Distribution Organization Structure



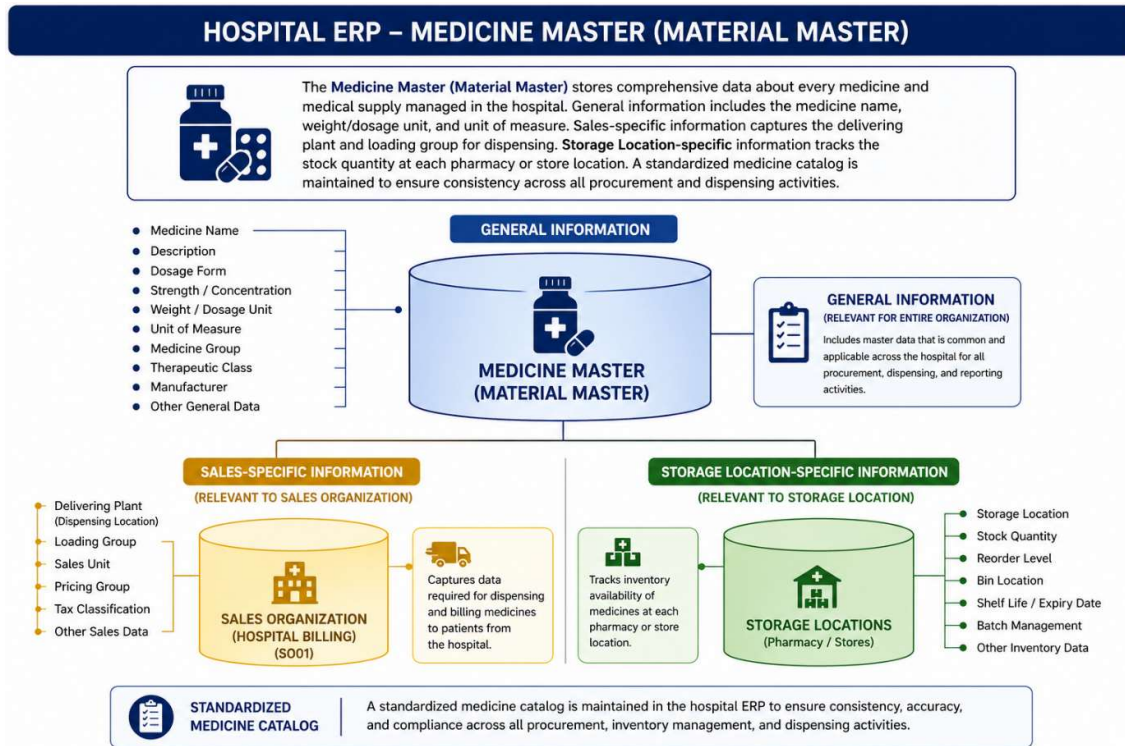
Master Data Required



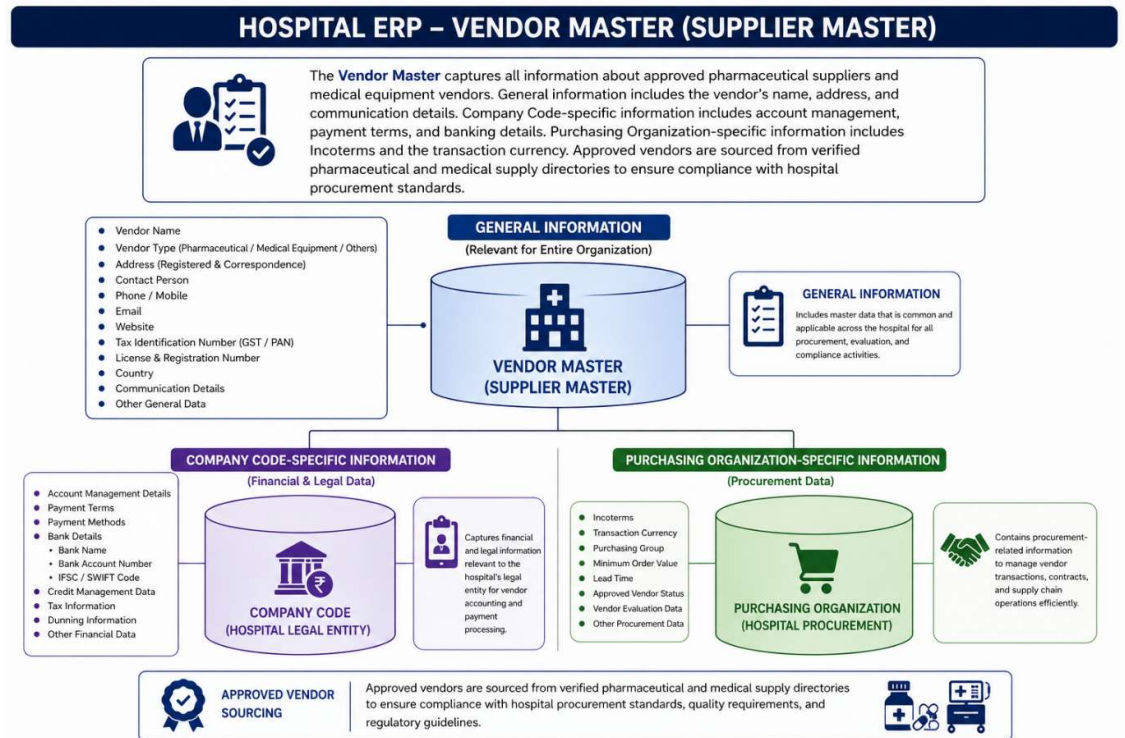
Patient Master (Customer Master)



Medicine Master (Material Master)



Vendor Master



Material Management (MM)

The MM module manages procurement and inventory of all medicines and medical equipment — from vendor selection to final goods receipt and stock management.

- **Create Vendor:** Register approved pharmaceutical and medical equipment suppliers in the system
- **Create Material:** Define medicine and medical supply records with dosage, unit of measure, and storage details
- **Display Stock:** View current inventory levels across pharmacy and equipment stores
- **Create Purchase Requisition:** Initiate a request to procure medicines or equipment when stock falls below threshold
- **Create RFQ:** Send Request for Quotation to shortlisted vendors for the required medicines
- **Create Quotation from Vendor:** Record vendor price quotes and delivery terms in the system
- **Price Based Bid Evaluation:** Compare vendor quotations and select the most cost-effective and compliant supplier
- **Create Purchase Order Referencing an RFQ:** Generate a formal purchase order to the selected vendor
- **Post Goods Receipt for Purchase Order:** Record the receipt of medicines/equipment into hospital inventory
- **Create and Post the First Supplier Invoice:** Process the vendor invoice against the purchase order
- **Post Outgoing Payment:** Execute payment to the vendor through the financial system
- **Check Stock:** Verify updated inventory levels after goods receipt and dispensing

Sales and Distribution (SD)

In the hospital ERP, patients are treated as customers and all hospital services — consultations, treatments, and procedures — are managed as service items within the SD module.

- **Create Customer (Patient):** Register a new patient as a customer in the SAP system with personal and billing details
- **Create a Contact Person:** Add a next-of-kin or authorized contact person linked to the patient record
- **Create Sales Inquiry:** Capture the patient's initial request for hospital services or consultation
- **Create Sales Quotation:** Generate a service quotation detailing estimated treatment costs for the patient

- **Create Sales Order:** Formally record the patient's confirmed services and treatment plan in the system
- **Manage Outbound Deliveries:** Coordinate the delivery of services, medicines, or medical devices to the patient
- **Create Billing Documents:** Generate an itemized invoice for all services rendered to the patient

Production Planning (PP)

In the hospital context, PP is adapted for healthcare planning — focusing on lab test scheduling, diagnostic workflow management, and resource optimization to drive efficient hospital operations.

- **Change Material Master Record:** Update medicine or supply specifications as required
- **Change Routing:** Modify lab test workflow and diagnostic process sequences
- **Display Product Group:** View groups of related diagnostic services or treatment protocols
- **Create Sales and Operations Plan (SOP):** Plan lab test volumes and diagnostic service capacity
- **Transfer SOP to Demand Management:** Convert service plans into actionable demand requirements
- **Run MPS with MRP:** Execute Master Production Scheduling and Material Requirements Planning for lab consumables
- **Review Stock/Requirements List:** Analyze supply vs. demand for lab materials and medicines
- **Convert Planned Order into Production Order:** Confirm and activate a lab test or diagnostic procedure order
- **Receive Goods in Inventory:** Record receipt of lab materials and reagents
- **Issue Goods to Production Order:** Dispense lab materials and medicines to the confirmed order
- **Confirm Production Completion:** Mark the lab test or diagnostic procedure as completed
- **Settle Costs of Production Order:** Allocate and settle the costs of completed lab services to the appropriate cost center

Module Integration

The core strength of this hospital ERP system is seamless module integration — ensuring every department operates in harmony with real-time data sharing. A single patient interaction triggers automatic updates across multiple modules, eliminating manual re-entry and reducing errors.

End-to-End Patient Journey:

- Patient Registration (SD) → Patient record created as a customer
- Treatment Recording (SD) → Services documented as sales order items
- Lab Tests Planned (PP) → Diagnostic workflows scheduled and resources allocated
- Medicines Issued (MM) → Stock automatically reduced at the pharmacy storage location
- Billing Generated (SD) → Itemized invoice created for all services and medicines
- Payment Processed (FICO) → Revenue posting created automatically from SD billing
- Staff Managed (HR) → Staff involvement in patient care tracked and payroll processed

Cross-Module Integration Points:

- SD → FICO: Billing a patient service in SD automatically updates Financial Accounting with revenue
- MM → Inventory: Dispensing medicine to a patient automatically reduces stock levels in Materials Management
- PP → Lab Services: Patient lab orders in Production Planning trigger specific testing protocols automatically
- HR → Care: Staff assignments and shift management are tracked in coordination with patient care activities

Conclusion

The implementation of SAP ERP for this private hospital has significantly transformed its operational landscape by integrating core healthcare processes, improving data visibility, and enhancing decision-making across all departments. By replacing fragmented and manual systems with a centralized digital platform, the hospital has achieved greater efficiency, transparency, and control over its patient services, procurement, financial management, and human resources.

The five-module architecture — SD for patient services, MM for medicine procurement, PP for lab scheduling, FICO for financial management, and HR for staff administration — creates a cohesive, intelligent healthcare platform where each module functions not in isolation but as part of a unified system. The three foundational pillars of organizational structure, master data, and transaction data ensure data integrity, traceability, and seamless operations across every module.

The proposed transition to SAP Fiori further strengthens this digital journey by introducing a user-friendly, role-based, and mobile-enabled interface. This advancement is expected to streamline bedside clinical operations, reduce approval delays, and improve real-time data capture — ultimately leading to higher staff productivity and better patient outcomes.

Through this implementation, the hospital demonstrates that ERP systems are not merely administrative tools — they are strategic enablers of patient-centric care, operational excellence, and sustainable institutional growth. The hospital's commitment to continuous digital evolution through SAP solutions positions it as a model for technology-driven healthcare transformation.

PPT – Private Hospital Management System

Revolutionizing Healthcare

Our Integrated Hospital ERP System

A team-built ERP solution for private healthcare — integrating patient care, inventory, billing, and finance into one unified platform.

HOSPITAL ERP PROJECT



The Disconnected Hospital: A System in Chaos

The Problem

Independent departments in private healthcare create critical inefficiencies — no shared data, no coordination.

Consequences

Delayed patient care, medicine stockouts, billing errors, and financial mismanagement become the norm.

The Cost

Patient dissatisfaction, rising operational expenses, and medical errors caused by fragmented information.



Our Vision: A Unified Healthcare Ecosystem



Seamless Integration

Connect all hospital departments — from patient admission to final billing — into one cohesive system.



Automate Routine Tasks

Free up medical staff from administrative burdens so they can focus on patient-centric care.



Real-Time Data Sharing

Enable immediate, informed decision-making across all departments with live data access.



Operational Efficiency

Enhance accuracy and reduce redundancy across every hospital process and workflow.



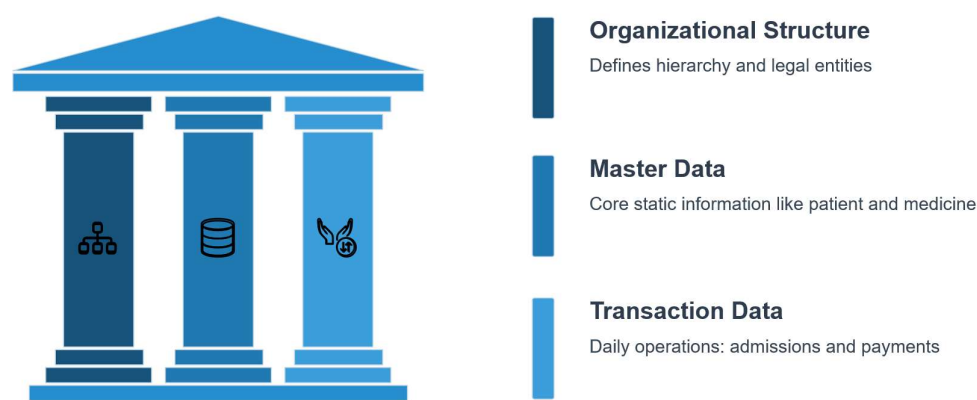
THE SOLUTION

The Heart of Our Solution

We implemented a **comprehensive, modular ERP system** designed specifically for private healthcare operations.

- Each module is specialized for its function — yet all feed into a **central, unified data pool**, creating a single source of truth for the entire hospital.

The ERP Blueprint: Three Foundational Pillars



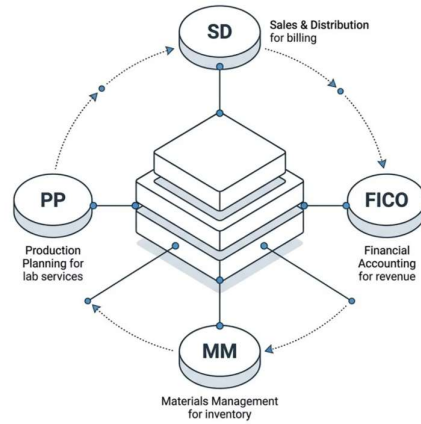
These three pillars form the backbone of our hospital ERP — ensuring data integrity, traceability, and seamless operations across every module.

Mapping the Hospital: Organizational Structure

01	02	03
Client	Company Code	Plant
The entire hospital network — the top-level container for all data.	The legal entity operating the hospital, handling financial reporting.	Represents the physical hospital location.
04	05	
Storage Location	Sales Organization	
Specific areas like the pharmacy or central sterile supply department.	The billing and patient accounts department.	

The Pillars of Information: Master Data in Action

Master data is the **static, foundational information** shared across all ERP modules. It ensures consistency and eliminates data redundancy hospital-wide.



Capturing Every Moment: Transaction Data at Work

Transaction data represents **daily operational events** — it leverages master data to execute and record every patient interaction.

Patient Registration

Capturing new patient details and admission information.

Treatment Recording

Documenting all medical interventions and procedures performed.

Medicine Usage

Tracking dispensing and consumption of pharmaceuticals in real time.

Billing & Payment

Generating accurate invoices and recording financial settlements.



The Power of Integration: One Action, Many Updates

The core strength of ERP is **automatic cross-module integration**. A single data entry triggers updates across multiple departments — eliminating manual re-entry and reducing errors.



SD → FICO

Billing a patient service in SD automatically updates Financial Accounting with revenue.

MM → Inventory

Dispensing medicine to a patient automatically reduces stock levels in Materials Management.

PP → Lab Services

Patient lab orders in Production Planning trigger specific testing protocols automatically.

Modules Utilized

Five core SAP ERP modules were deployed, each mapped to a critical hospital function — creating a comprehensive, end-to-end management system.



SD — Sales & Distribution

Patient services, appointments, and billing cycles.

MM — Materials Management

Medicine procurement, stock levels, and inventory control.

PP — Production Planning

Lab test scheduling and diagnostic workflow planning.

FICO — Finance & Controlling

Financial accounting, cost tracking, and reporting.

HR — Human Resources

Staff management, payroll, and workforce administration.

Streamlining Patient Journeys: The SD Module

Patients are treated as customers and hospital services — consultations, diagnoses, treatments — are managed as service items.

1

Patient Registration

Initiates the patient's complete journey within the ERP system.

2

Service Recording

Accurately documents all consultations, diagnoses, and treatments provided.

3

Billing & Invoice

Generates precise, itemized bills ensuring transparency and revenue accuracy.

☐ **Outcome:** Every service is correctly captured and billed — maximizing revenue and patient satisfaction.



MM MODULE

Ensuring Medicine & Equipment Availability

The MM module manages procurement and inventory of all medicines and medical equipment — from vendor selection to final goods receipt.

Vendor Selection

Identify and manage approved suppliers for medicines and equipment

Purchase & Receipt

Streamlined purchasing workflows with automated goods receipt

Auto Stock Updates

Inventory is automatically adjusted as medicines are issued to patients

PP MODULE

Healthcare Planning



PP is adapted for healthcare planning—focusing on scheduling, workflow management, and resource optimization to drive efficient hospital operations.

→ Lab Test Scheduling

Plan and track diagnostic tests across departments

→ Workflow Management

Coordinate hospital processes and patient flow

→ Resource Planning

Allocate equipment, rooms, and staff efficiently

FICO MODULE

Mastering Financial Operations

FICO handles all financial activities within the hospital — from recording transactions to generating comprehensive financial reports.



Transaction Recording

Every financial event is captured accurately in real time



Payments Management

Track and manage all incoming and outgoing payments



Revenue & Expenses

Monitor financial performance across departments



Financial Reporting

Generate reports for compliance and decision-making

📌 **Integration Point:** Billing generated in SD automatically updates FICO — ensuring financial records are always current.

HR MODULE

Empowering Hospital Staff

Workforce Management

The HR module manages all aspects of hospital staff — ensuring the right people are in the right roles at the right time.



Employee Records

Centralized profiles for all hospital staff

Attendance & Payroll

Automated tracking and compensation processing

Role Management

Define access levels and responsibilities per role

MODULE INTEGRATION

The Power of Synergy

The core strength of our ERP system is seamless module integration — ensuring every department operates in harmony with real-time data sharing.

SD → FICO

Billing automatically updates financial records

MM → Inventory

Stock updates in real time as medicines are used

PP → Operations

Schedules services and coordinates workflows

HR → Care

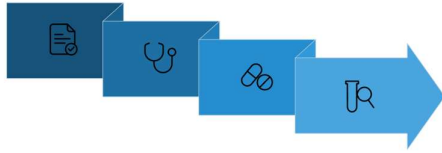
Tracks staff involvement in patient care



END-TO-END PROCESS

From Patient Arrival to Recovery

Every step of the patient journey is supported by integrated ERP modules working together in real time.



“When a patient arrives:

- 1.Patient registration (SD)
- 2.Treatment
- 3.Lab tests planned (PP)
- 4.Medicines issued (MM)
- 5.Billing generated (SD)
- 6.Payment processed (FICO)
- 7.Staff managed (HR)

All modules work together in real time.”

All modules communicate in real time — creating a unified, error-free patient experience from registration to discharge.

KEY ADVANTAGES

Transforming Healthcare Delivery



Real-Time Data

Immediate insights across all departments



Reduced Errors

Automation eliminates manual data entry mistakes



Better Coordination

Seamless communication across all hospital units



Efficient Inventory

Automated stock tracking prevents shortages



Improved Patient Service

Streamlined processes enhance the patient experience



Conclusion

"In our project, we focused not just on modules, but on how **master data**, **organizational structure**, and **integration** together form a complete ERP system."

Together, these three pillars ensure that every module — MM, PP, FICO, HR, and SD — functions not in isolation, but as part of a cohesive, intelligent healthcare platform.